CODE OF BOOTH MULTIPLIER

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/booth/booth_mul.vhd [VHDL]

```vhdl
----------------------------------------------------------------------------------
-- Company:
-- Engineer:
--
-- Create Date: 09/10/2026 04:26:32 PM
-- Design Name:
-- Module Name: booth_mul - Behavioral
-- Project Name:
-- Target Devices:
-- Tool Versions:
-- Description:
--
-- Dependencies:
--
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
--
----------------------------------------------------------------------------------
LIBRARY IEEE;
USE IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
USE IEEE.NUMERIC_STD.ALL;
--library UNISIM;
--use UNISIM.VComponents.all;

ENTITY booth_mul IS
    PORT (
        a : IN signed (3 DOWNTO 0);
        b : IN signed (3 DOWNTO 0);
        p : OUT signed (7 DOWNTO 0));
END booth_mul;

ARCHITECTURE Behavioral OF booth_mul IS
BEGIN

    PROCESS (a, b)
        VARIABLE q_pair : STD_LOGIC_VECTOR(1 DOWNTO 0);
        VARIABLE acc : signed (3 DOWNTO 0);
        VARIABLE q : signed (3 DOWNTO 0);
        VARIABLE total : STD_LOGIC_VECTOR (8 DOWNTO 0);

        VARIABLE w : INTEGER := 3;
```

```
46
47      BEGIN
48          q_pair := "00";
49          acc := "0000";
50          q := b;
51          total := STD_LOGIC_VECTOR(acc) & STD_LOGIC_VECTOR(q) & q_pair(0);
52
53          FOR i IN 0 TO w LOOP
54
55              q_pair := total(1 DOWNTO 0);
56
57              CASE(q_pair) IS
58                  WHEN "01" => acc := acc + a;
59                  WHEN "10" => acc := acc - a;
60                  WHEN OTHERS => NULL;
61              END CASE;
62
63              total := STD_LOGIC_VECTOR(acc) & STD_LOGIC_VECTOR(q) & q_pair(0);
64              total := total(8) & total(8 DOWNTO 1);
65              acc := signed(total(8 DOWNTO 5));
66              q := signed(total(4 DOWNTO 1));
67
68          END LOOP;
69
70          p <= signed(total(8 DOWNTO 1));
71
72      END PROCESS;
73
74  END Behavioral;
```

CODE OF TEST BENCH FOR BOOTH MULTIPLIER

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/tb/booth/booth_mul_tb.vhd [VHDL]

```
1  ----------------------------------------------------------------------------------
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date: 09/10/2026 04:26:32 PM
6  -- Design Name:
7  -- Module Name: booth_mul_tb - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
```

```
-- Additional Comments:
--
----------------------------------------------------------------------------------
LIBRARY IEEE;
USE IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
USE IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

ENTITY booth_mul_tb IS
    --  Port ( );
END booth_mul_tb;

ARCHITECTURE testing OF booth_mul_tb IS
    COMPONENT booth_mul IS
        PORT (
            a : IN signed (3 DOWNTO 0);
            b : IN signed (3 DOWNTO 0);
            p : OUT signed (7 DOWNTO 0));
    END COMPONENT;
    SIGNAL a : signed(3 DOWNTO 0);
    SIGNAL b : signed(3 DOWNTO 0);
    SIGNAL p : signed(7 DOWNTO 0);

BEGIN
    booth_multiplier_test : booth_mul PORT MAP(
        a => a,
        b => b,
        p => p
    );
    PROCESS

    BEGIN
        a <= "0101";
        b <= "1010";

        WAIT FOR 10ns;

        a <= "0011";
        b <= "1100";

        WAIT FOR 10ns;

```

```
        a <= "1010";
        b <= "1100";

        WAIT FOR 10ns;

        a <= "0001";
        b <= "1000";

        WAIT FOR 10ns;

        a <= "1110";
        b <= "1011";

        WAIT FOR 10ns;

        a <= "1111";
        b <= "1111";

        WAIT FOR 10ns;

        a <= "0000";
        b <= "0000";

        WAIT FOR 10ns;

        a <= "1010";
        b <= "1111";

        WAIT FOR 10ns;

        a <= "0001";
        b <= "1111";

        WAIT FOR 10ns;

        a <= "1010";
        b <= "1111";
        WAIT;

    END PROCESS;

END testing;
```

SCREENSHOT OF CODE

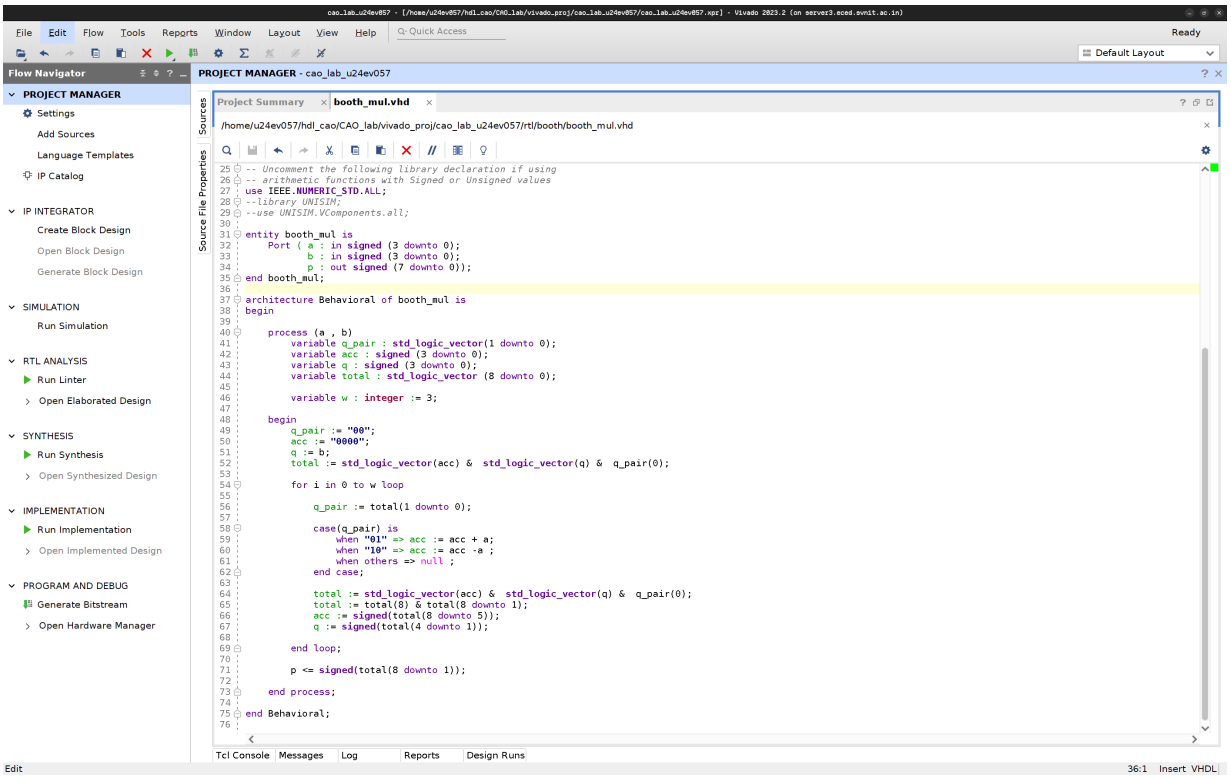


Figure 1: Design

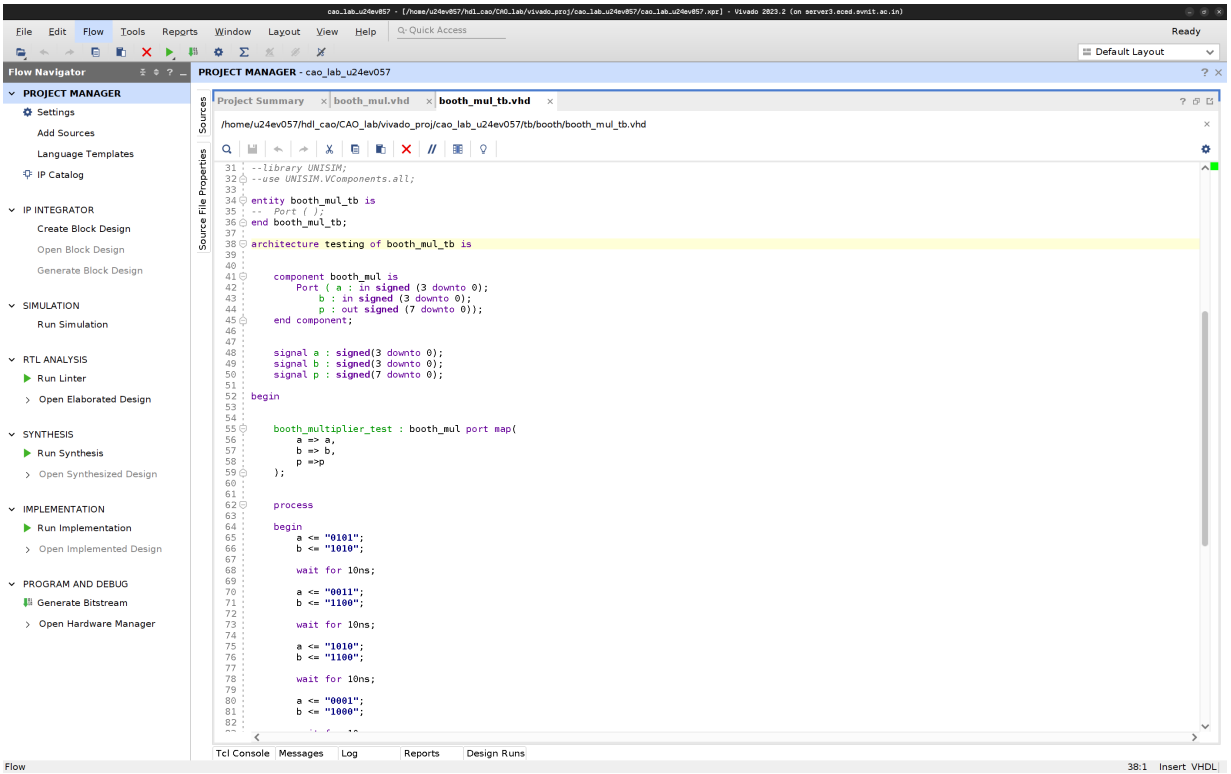


Figure 2: Tb

WAVEFORM

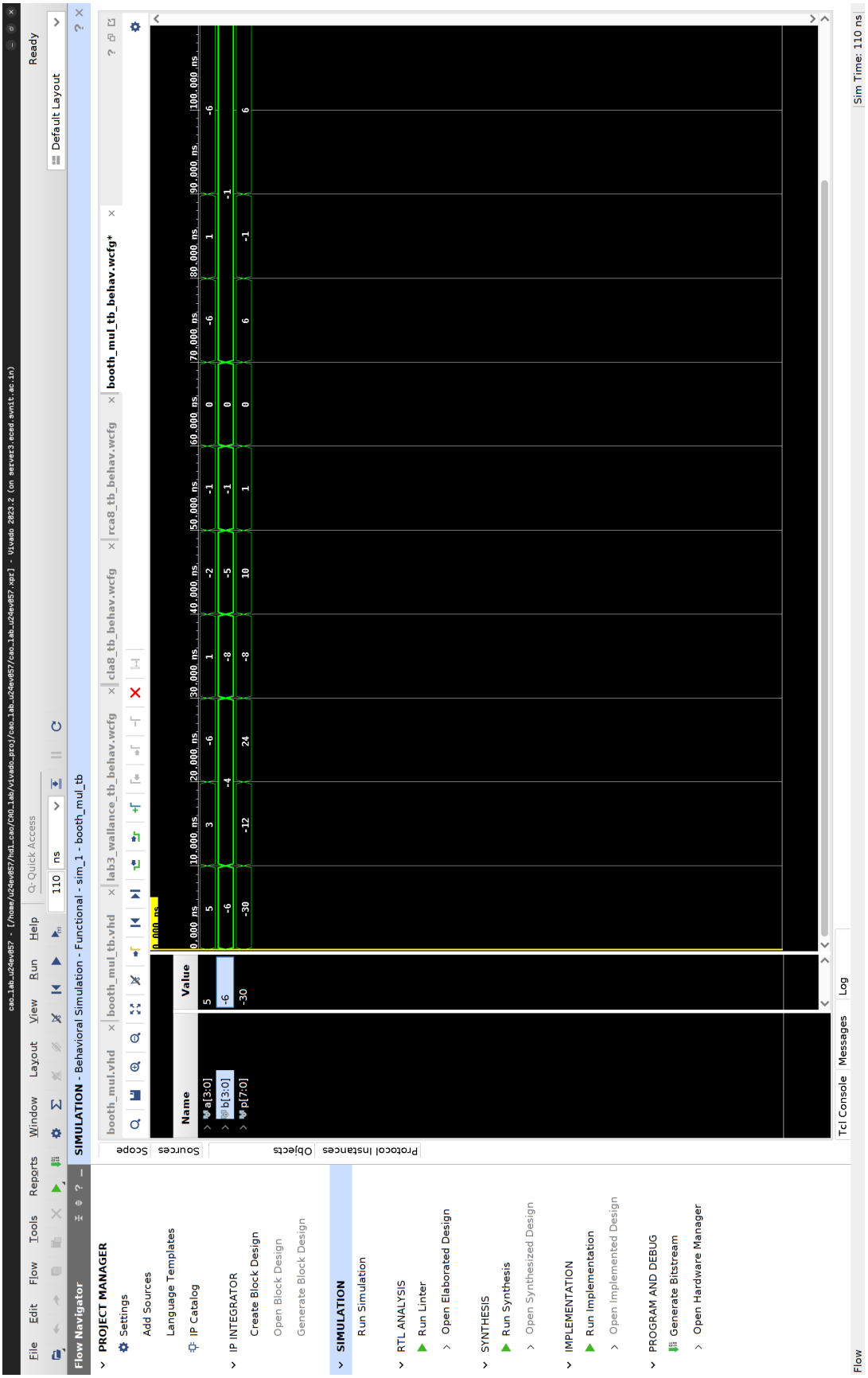


Figure 3: Waveform